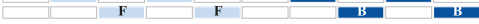


(a)

Strategy	8-GPU assignment	Pipeline schedule	Observed outcome
FSDP-only failed	 D: data-parallel shards	DP shard 	Not feasible 1/1 attempts failed OOM during FSDP buffer init
PP2 + FSDP valid	 S0: 4 GPUs S1: 4 GPUs	Stage 0  Stage 1 	7,126 tokens/s 95% CI +/- 12.3, n=3 2,299 ms/step, 27.9 GiB highest throughput
PP4 + FSDP valid	 S0: 2 GPUs S1: 2 GPUs S2: 2 GPUs S3: 2 GPUs	Stage 0  Stage 1  Stage 2  Stage 3 	5,786 tokens/s 95% CI +/- 7.7, n=3 1,416 ms/step, 24.9 GiB lower memory, lower throughput
PP2 + VPP2 + FSDP failed	 S0: 4 GPUs S1: 4 GPUs	S0-V0  S0-V1  S1-V0  S1-V1 	Not feasible 4/4 attempts failed GBS=4 illegal; GBS=8 OOM

forward

backward

pipeline bubble / idle slot

